

Rahul Nadkarni

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RESEARCH INTERESTS

I am interested in work that takes a rigorous scientific approach to developing language model-based systems that are useful, safe, and reliable. My most recent work involved understanding how LMs' training data shapes their knowledge capabilities, and I am broadly interested in better understanding and curating training data to optimize desired model behaviors. I am particularly interested in applications of LMs to high-impact domains that affect human well-being, and generally in building capable LMs to assist humans in technical applications.

EDUCATION

University of Washington – Seattle, WA

2026 – *Ph.D. Computer Science & Engineering* (Advisor: Noah A. Smith)

2017 – *M.S. Computer Science & Engineering*

University of California, Berkeley – Berkeley, CA

2015 – *B.S. Electrical Engineering & Computer Science / Bioengineering (double major)*

RESEARCH EXPERIENCE

Graduate Student Researcher

University of Washington – Seattle, WA

- 2020–2026 Various research projects in natural language processing, including:
 - Adapting domain-specific language models for scientific knowledge base completion
 - Exploring methods for predicting performance of instruction-tuned language models
 - Doing interventions on LLM pretraining by altering data and repeating pretraining runs
 - Testing parameter-efficient fine-tuning (PEFT) methods for domain adaptation
 - Exploring merging / ensembling of instruction-tuned LMs for modularity and performance
 - Contributed to developing an LM tool use and function calling setup for question answering
- 2015–2020 Various projects in ML for time series applied to neuroscience, including:
 - Developing a graphical LASSO-type model for time series to learn brain network structure
 - Applying time-varying SSMs to neuroimaging data to estimate dynamic brain connectivity

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Machine Learning (Ph.D.)

June – September 2021

Facebook – Seattle, WA

Received full-time offer

Content Integrity team, worked on improving classifiers for detecting objectionable online content.

Software Engineering Intern, Ph.D.

June – September 2017

Google – Seattle, WA

Received returning internship offer

Maps team, worked on visualizing traffic data and adding to the traffic server backend.

SELECTED PUBLICATIONS

Natural Language Processing

Rahul Nadkarni, Yanai Elazar*, Hila Gonen*, Noah A. Smith. [Rewriting History: A Recipe for Interventional Analyses to Study Data Effects on Model Behavior](#). **TACL** (2026).

Rahul Nadkarni, Yizhong Wang, Noah A. Smith. [Third-Party Language Model Performance Prediction from Instruction](#). **arXiv** (2024).

Zhoujun Cheng, Tianbao Xie, Peng Shi, Chengzu Li, **Rahul Nadkarni**, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, Tao Yu. [Binding Language Models in Symbolic Languages](#). **ICLR** (2023).

Rahul Nadkarni, David Wadden, Iz Beltagy, Noah A. Smith, Hannaneh Hajizhirshi, Tom Hope. [Scientific Language Models for Biomedical Knowledge Base Completion: An Empirical Study](#). **AKBC** (2021).

SELECTED
PUBLICATIONS

Machine Learning and Neuroscience

Jordan Drew, Nicholas Foti, **Rahul Nadkarni**, Eric Larson, Emily Fox, Adrian KC Lee. [Using a linear dynamic system to measure functional connectivity from M/EEG](#). **Journal of Neural Engineering**, **21** (2024).

Jordan Drew, Eric Larson, Nicholas Foti, **Rahul Nadkarni**, Emily Fox, Adrian KC Lee. [Dynamic functional connectivity in auditory attention task](#). **The Journal of the Acoustical Society of America** (2021).

Rahul Nadkarni, Nicholas J. Foti, Adrian KC Lee, and Emily B. Fox. [A hierarchical state-space model with Gaussian process dynamics for functional connectivity estimation](#). **NeurIPS Workshop on Learning Meaningful Representations of Life** (2019).

Rahul Nadkarni, Nicholas J. Foti, and Emily B. Fox. [Learning dynamic functional connectivity networks from infant magnetoencephalography data](#). **NeurIPS BigNeuro Workshop** (2017).

Nicholas J. Foti, **Rahul Nadkarni**, Adrian KC Lee, and Emily B. Fox. [Sparse plus low-rank graphical models of time series for functional connectivity in MEG](#). **SIGKDD Workshop on Mining and Learning from Time Series** (2016).

SKILLS

Languages: Python (proficient); BASH scripting, Java, C/C++ (working knowledge)

Libraries/Frameworks: PyTorch, HuggingFace, scikit-learn, wandb, NumPy, SciPy

Systems/Infra: Slurm, AWS (EC2), Hadoop

Other: Git/Github, Conda, OpenAI API, L^AT_EX

GRADUATE
COURSEWORK

Natural Language Processing	Computer Vision	Machine Learning for Big Data
Graphical Models	Statistical Inference	Convex Optimization
Online Learning	Algorithms	Database Systems

TEACHING
EXPERIENCE

University of Washington – Seattle, WA

Teaching Assistant, Introduction to AI	March – June 2016, September – December 2020
Teaching Assistant, Data Structures and Algorithms	September 2015 – March 2016

HONORS &
AWARDS

NSF IGERT Fellowship in Big Data and Data Science	August 2017 – September 2019
Honorable Mention, NSF Graduate Research Fellowship Program (GRFP)	Awarded March 2017
Dean’s Honors List, UC Berkeley	January 2012 – May 2015
Member, Eta Kappa Nu IEEE Honor Society, UC Berkeley (Mu Chapter)	January 2013